

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

Misura base Z : $p_0 = |\langle 0|\psi\rangle|^2 = |\alpha\langle 0|0\rangle + \beta\langle 0|1\rangle|^2 = |\alpha|^2$
 $p_1 = |\beta|^2 = 1 - |\alpha|^2$

$$\langle Z \rangle = \langle \psi|Z|\psi \rangle = \langle \psi|(|0\rangle\langle 0| - |1\rangle\langle 1|)|\psi \rangle = \langle \psi|0\rangle\langle 0|\psi \rangle - \langle \psi|1\rangle\langle 1|\psi \rangle =$$

$$= |\langle 0|\psi\rangle|^2 - |\langle 1|\psi\rangle|^2 = p_0 - p_1$$

Misura base X : Applico H e poi misuro

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$H|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} \alpha + \beta \\ \alpha - \beta \end{pmatrix} \cdot \frac{1}{\sqrt{2}} = |\psi_x\rangle$$

$$= \frac{\alpha + \beta}{\sqrt{2}} |0\rangle + \frac{\alpha - \beta}{\sqrt{2}} |1\rangle$$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

$$p_0 = |\langle 0|\psi_x\rangle|^2 = \left| \frac{\alpha + \beta}{\sqrt{2}} \langle 0|0\rangle + \frac{\alpha - \beta}{\sqrt{2}} \langle 0|1\rangle \right|^2 = \frac{|\alpha + \beta|^2}{2} = \frac{|\alpha|^2 + |\beta|^2 + \alpha\beta^* + \alpha^*\beta}{2}$$

$$p_1 = |\langle 1|\psi_x\rangle|^2 = \frac{|\alpha - \beta|^2}{2} = \frac{|\alpha|^2 + |\beta|^2 - \alpha^*\beta - \alpha\beta^*}{2}$$

$$\langle X \rangle = \langle \psi|X|\psi \rangle = \langle \psi|(|0\rangle\langle 1| + |1\rangle\langle 0|)|\psi \rangle = \quad \underline{|\psi\rangle = \alpha|0\rangle + \beta|1\rangle}$$

$$= \langle \psi|0\rangle\langle 1|\psi \rangle + \langle \psi|1\rangle\langle 0|\psi \rangle = \alpha^*\beta + \alpha\beta^*$$

$$(\alpha^*\langle 0| + \beta^*\langle 1|)|0\rangle = \alpha^* \quad \langle 1|(\alpha|0\rangle + \beta|1\rangle) = \beta$$

$$p_0 - p_1 = \alpha^*\beta + \alpha\beta^* = \langle X \rangle = 2 \operatorname{Re}(\alpha\beta^*)$$